Omer El Idrissi

Systems Programmer, Linux Contributor

omer.e.idrissi@gmail.com +387644390489 Zavidovići, Federation of Bosnia and Herzegovina

github.com/omeridrissi/

Projects

cachepix
cachepix is a high-performance C library for processing PPM (portable pixmap) images, with a strong focus on data layout, cache efficiency, and SIMD optimization.

fifine_mic_driver
This was my first attempt at making a high-performance linux driver for a physical device that i own. It communicates with the microphone's USB isochronous endpoint and efficiently delivers the pure uncompressed stream with a fast single-consumer, multiple producers model of ring buffer processing.

prongc
It's a static C cross-function variable dependency analysis tool that checks if two or more functions touch the same data. It supports local variables, parameters, static and global variables. It builds a graph structure the represents the collected function info and all variable accesses of that function and it's callees and checks if they touch the same variables. It uses libclang to parse the source code and build the ASTs.

Skills

C Programming

Linux

Software automation

Core Competencies
- Systems Programming
- Fast and Efficient Algorithm Design

Familiarity with Jenkins

Open Source Contributions

Linux kernel - four accepted patches
- [staging]: Cleaned up vendor-defined error macros, replacing with proper kernel errno codes (2-patch series)
- [staging]: Fixed style mistake and minor type fix

Education

Self-directed study - Systems Programming and Kernel Development
2023 – Present

Performance & Architecture:
- Studied memory hierarchy, caching, and code optimization from Computer Systems: A Programmer's Perspective (Bryant & O'Hallaron), directly applied in SIMD and cache-line-aware design for cachepix

Hardware & Electronics Basics:
- Worked through Practical Electronics for Inventors (Scherz & Monk), developed ability to read and interpret chip datasheets and USB device specifications

Kernel Development:
- Studied kernel module architecture and USB subsystem fundamentals through linux kernel documentation and the Linux Kernel Module Programming Guide, applied in building fifine_mic_driver and analyzing usb-skeleton.c

Continuous learning:
- Routine use of primary kernel documentation and community-created manpages for ongoing development
- Effective at leveraging AI-assisted research combined with source-level verification against actual kernel code.